

Dietary Fatty Acids and Pancreatic Cancer in the NIH-AARP Diet and Health Study



Background Previous research relating dietary fat, a modifiable risk factor, to pancreatic cancer has been inconclusive. **Methods** We prospectively analyzed the association between intakes of fat, fat subtypes, and fat food sources and exocrine pancreatic cancer in the National Institutes of Health–AARP Diet and Health Study, a US cohort of 308 736 men and 216 737 women who completed a 124-item food frequency questionnaire in 1995–1996. Hazard ratios (HRs) and 95% confidence intervals (CIs) were calculated using Cox proportional hazards regression models, with adjustment for energy intake, smoking history, body mass index, and diabetes. Statistical tests were two-sided. **Results** Over an average follow-up of 6.3 years, 865 men and 472 women were diagnosed with exocrine pancreatic cancer (45.0 and 34.5 cases per 100 000 person-years, respectively). After multivariable adjustment and combination of data for men and women, pancreatic cancer risk was directly related to the intakes of total fat (highest vs lowest quintile, 46.8 vs 33.2 cases per 100 000 person-years, HR = 1.23, 95% CI = 1.03 to 1.46; $P_{\text{trend}} = .03$), saturated fat (51.5 vs 33.1 cases per 100 000 person-years, HR = 1.36, 95% CI = 1.14 to 1.62; $P_{\text{trend}} < .001$), and monounsaturated fat (46.2 vs 32.9 cases per 100 000 person-years, HR = 1.22, 95% CI = 1.02 to 1.46; $P_{\text{trend}} = .05$) but not polyunsaturated fat. The associations were strongest for saturated fat from animal food sources (52.0 vs 32.2 cases per 100 000 person-years, HR = 1.43, 95% CI = 1.20 to 1.70; $P_{\text{trend}} < .001$); specifically, intakes from red meat and dairy products were both statistically significantly associated with increased pancreatic cancer risk (HR = 1.27 and 1.19, respectively). **Conclusion** In this large prospective cohort with a wide range of intakes, dietary fat of animal origin was associated with increased pancreatic cancer risk.